

Predicting NBA shot make probability

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Approach

I trained a histogram-based gradient boosting classifier on 425,719 labeled, non-fouled shots supplied in the repository. It outputs a probability of a make for each shot. The model learns nonlinear effects and interactions, such as the different meaning of a defender being two feet away on a dunk versus a jumper. I used only the supplied data.

Inputs include shot distance, type, court location, three-point status, dribbles, shot clock, game state, shooter speed, and contest information. The four tracking distances for the closest defender in the second before release were parsed into numeric features; I also calculated changes in that distance as the shot approached. Missing contest distances remain missing. The 180 most frequent shooters and 100 most frequent closest defenders retain separate categorical identities; others are grouped as OTHER. Team, season, and month identifiers are included as categories. Shot ID and the target are excluded from model inputs.

Validation and output

I held out a stratified random 20% of the labeled shots (85,144 attempts) before fitting the validation model. A constant prediction equal to the training make rate scored **0.689627** log-loss; the model scored **0.628205** on my computer. Lower is better. After validation, I refit the same configuration on all labeled shots and wrote probabilities for all 213,977 test shots, preserving the template's columns and shot order. The script checks both before writing.

The random split estimates performance on shots resembling the pooled sample, including familiar shooters and seasons. It does not establish performance on an unseen season or wholly new players. The held-out metric uses real outcomes; the test labels and the organization's in-house reference probabilities are unavailable to me.

What drives the predictions

I measured permutation importance on 10,000 validation shots with two shuffles per feature. Each bar is the increase in log-loss when that feature is scrambled while the others are held fixed. Distance dominates; shot type and contest proximity add useful information. Individual importances can understate the combined value of correlated defender and location fields.

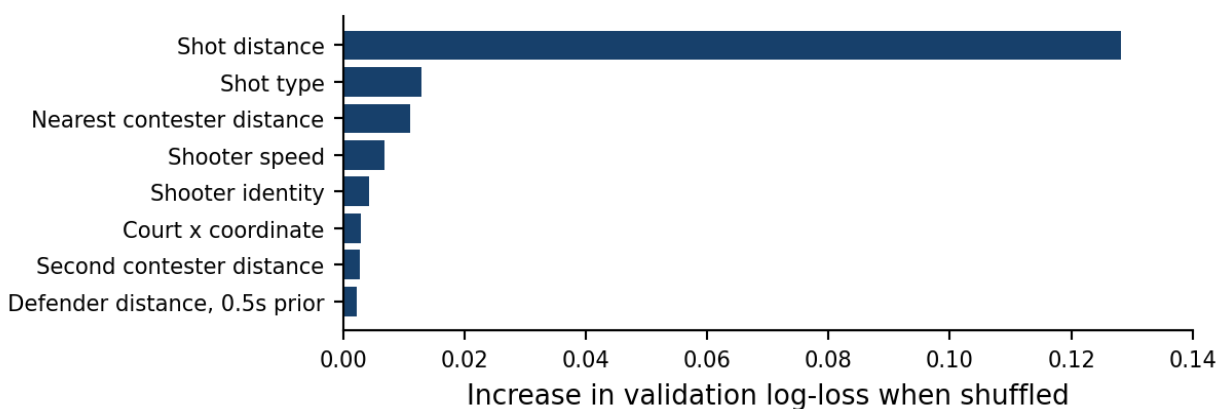


Figure 1. Validation permutation importance. The largest eight increases are shown; values below about 0.003 should be read cautiously.

Probability checks

I grouped validation predictions into 0.1-wide probability bins. The observed make rate generally follows the predicted rate. In the 0.7-0.8 bin, the mean prediction was 0.747 versus an observed 0.724 across 2,918 shots; this suggests some local overconfidence. The lowest bin contains only 118 shots, so its estimate is noisy.

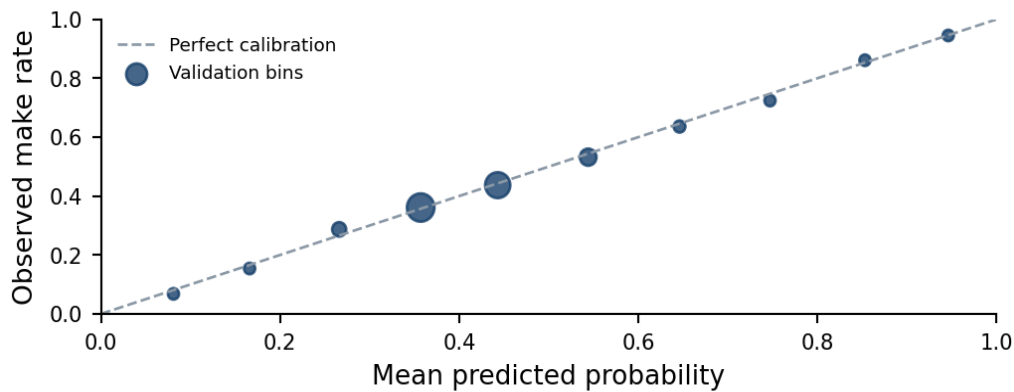


Figure 2. Mean predicted probability versus observed make rate in validation bins; point area indicates approximate bin size.

Next steps and limitations

With more time, I would check stability by holding out an entire season and evaluate log-loss by shot type and shooter frequency. I would compare a simpler distance-and-shot-type model with this fuller specification, test probability calibration on a separate calibration split, and examine whether tracking trajectories add value after accounting for release-time distance. A richer trajectory or defender-angle feature could help distinguish active contests from nearby but ineffective defenders. These are proposals, not claims that they improve the current result.